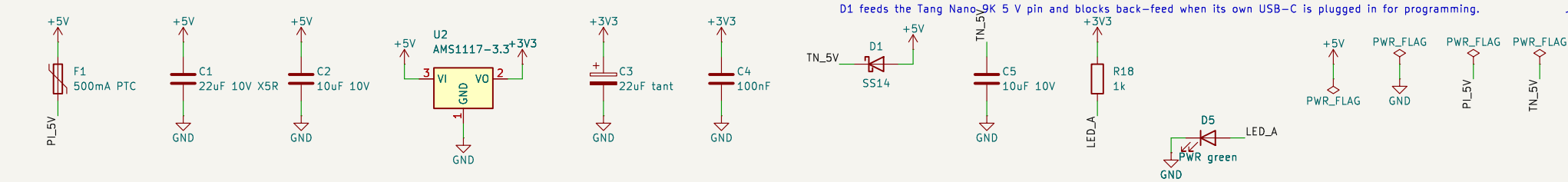


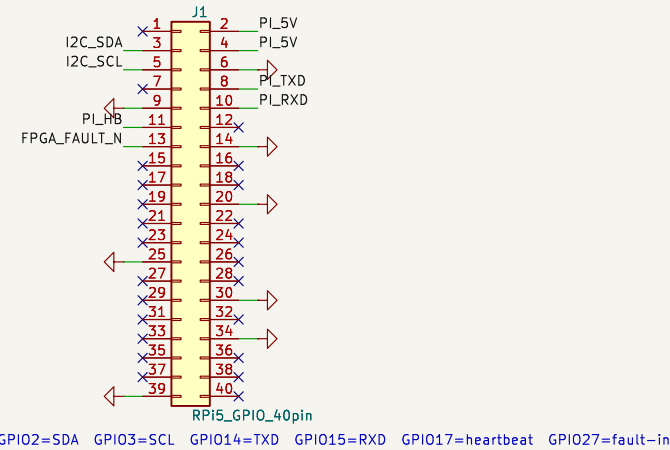
POWER

Pi 5 is powered by a USB-C power bank; this board takes 5 V from the Pi 40-pin header (pins 2/4) through a 500 mA PTC (protects the Pi 5 V rail). AMS1117 makes +3V3 for sensors, motors, IMU, buttons.



J1 RASPBERRY PI 5 40-PIN HEADER

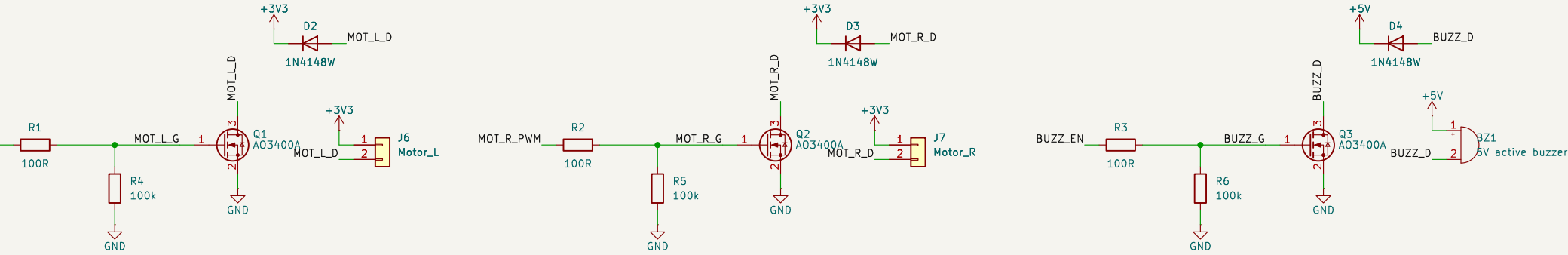
2x20 pin header + 40-way IDC ribbon to the Pi GPIO header. Board = Pi 5 outline (85 x 56 mm), same mounting holes, stacked on 20 mm standoffs above the Pi (clears USB/Ethernet jacks and the active cooler). Pi 3V3 (pins 1, 17) intentionally NOT connected: this board has its own 3.3 V regulator.



GPIO2=SDA GPIO3=SCL GPIO14=TXD GPIO15=RXD GPIO17=heartbeat GPIO27=fault-in

HAPTICS & BUZZER

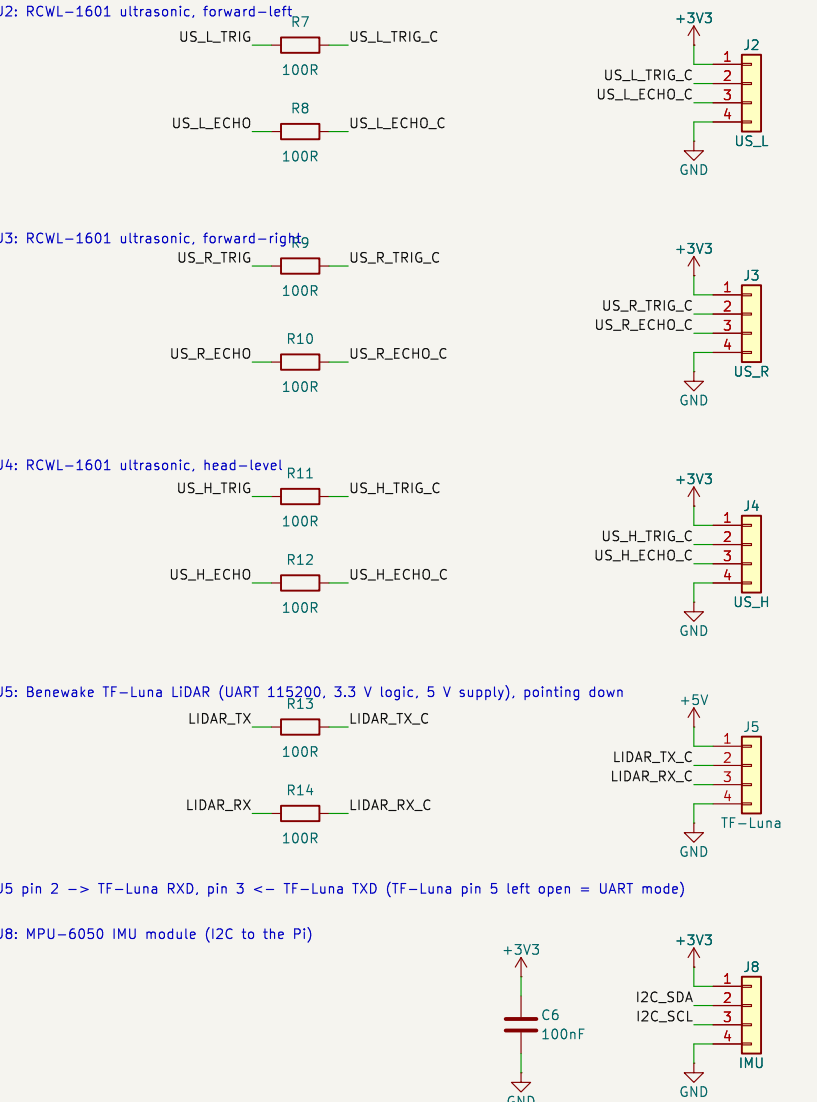
A03400A logic-level N-MOSFETs (Vgs(th) <= 1.45 V) switched by FPGA 3.3 V PWM. 100k gate pull-downs keep outputs OFF while the FPGA boots. 1N4148W diodes clamp motor/buzzer flyback. 3 V coin ERM's run from +3V3.



J6/J7 go to the coin motors on the left/right harness straps; BZ1 sounds the "check device" alarm.

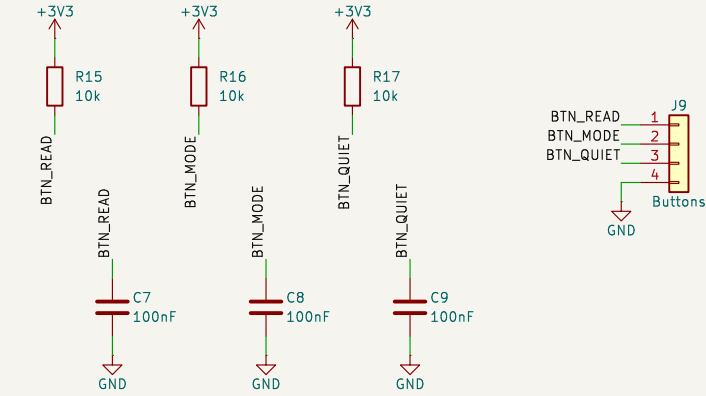
SENSORS (off-board, JST-XH)

100R series resistors limit fault current and ringing on the sensor cables.



BUTTONS (READ = OCR, MODE, QUIET)

10k pull-up + 100nF = 1 ms RC filter; FPGA also debounces digitally.



Design stage – not yet fabricated. Verify Tang Nano 9K row spacing before ordering.
FPGA safety island (Tang Nano 9K) + sensor / haptic interface for the Pi 5 AI path
Team SVNIT – Vishwakarma Awards 2026–27

Sheet: /
File: VisionAid_Carrier.kicad_sch

Title: VisionAid Carrier Board (stacks on Raspberry Pi 5, 40-pin ribbon)

Size: A3 Date: 2026-10-08 Rev: v0.1
KiCad E.D.A. 9.0.8 Id: 1/1